

Charge-and-Release Rhythms in Ore-Forming Systems Are Scalar Clocks, Not Level Repulsion: A Cautionary Random-Matrix Analysis of Metallogenic Timing

Version 2.0 (erratum). A single ore system is a relaxation oscillator—a scalar renewal clock—structurally not a GOE spectrum

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Abstract

This is version 2.0, a self-correcting erratum. Version 1 of this work claimed that the multi-stage timing of single ore-forming systems carries the Gaussian-orthogonal/unitary-ensemble (GOE/GUE) *level-repulsion* signature of random matrix theory (RMT), interpreted as a charge-and-release “memory.” That central claim was wrong, and we retract it here. Re-auditing the identical data with the diagnostics the original analysis omitted—the shuffle (permutation) test, the coefficient of variation (CV), an i.i.d. renewal-clock comparison, and an age-error Monte Carlo—shows that the high spacing ratios are a *scalar-clock* (narrow-marginal renewal) signature, not sequential level repulsion. The pooled spacing ratios are reproduced *exactly* ($\langle r \rangle = 0.60$ Andean, 0.57 Nanling, 0.71 Tethyan, 0.68 orogenic gold), but (i) they are invariant under shuffling, so they reflect the marginal spacing distribution, not the sequence; (ii) a zero-repulsion gamma renewal clock with the same CV reproduces every value; (iii) the small negative “rigidity” ($\text{obs} - \text{shuf} = -0.031$ for gold) is fully reproduced by pushing a *zero-repulsion* null through the same within-deposit mean-normalization—it is a closed-data (compositional) artifact, not spectral rigidity; and (iv) within the quoted geochronological errors the pooled $\langle r \rangle$ ranges across the entire Poisson-to-GUE axis, so the point estimate is unresolvable at these sample sizes ($n = 21\text{--}40$, far below the $n \gtrsim 300$ needed to distinguish a scalar clock from GOE)—and

the $\pm 1\text{--}5$ Ma analytical errors of the underlying chronometers are themselves comparable to the intervals, so isotopic dating cannot in principle resolve the sequential rigidity a Wigner–Dyson spectrum would require. A charge-and-release reservoir is a classical relaxation oscillator, i.e. a renewal process; by the Bohigas–Giannoni–Schmit criterion it is structurally *not* a chaotic GOE spectrum. **What survives is the superposition result:** pooling many independent sources drives the local statistic toward Poisson ($\langle r \rangle \rightarrow 0.39$), a genuine and retained finding. We keep the four-province dataset and the superposition analysis, retract the level-repulsion interpretation, and reframe single ore systems as scalar charge-release clocks. Basis of the correction: the scalar-clock audit note, Zenodo <https://doi.org/10.5281/zenodo.20980670>.

Keywords: random matrix theory; scalar clock; renewal process; relaxation oscillator; metallogeny; ore geochronology; spectral superposition; self-correction

Code/data: <https://github.com/Ruqing1963/metallogeny-rmt>

1 Changes from version 1 (erratum)

This section states plainly what changed and why, so that readers of version 1 can see exactly which claims are withdrawn.

- **Retracted.** The claim that single ore-forming systems exhibit GOE/GUE *level repulsion*, and the associated “intrinsic charge-release memory” and “grand unification of five deep-Earth systems” framing. The non-magmatic orogenic-gold “control” does *not* establish intrinsic repulsion; like the magmatic provinces it is a scalar clock.
- **Why.** The spacing ratio $\langle r \rangle$ is two-ended: a high value can come either from genuine sequential level repulsion *or* from a narrow marginal (scalar) spacing distribution. Version 1 read a high $\langle r \rangle$ as repulsion without running the shuffle test that separates the two. A charge-and-release reservoir is a relaxation oscillator—a renewal process with a characteristic recurrence time—whose narrow interval distribution produces a high $\langle r \rangle$ with *no* sequential correlation. By the Bohigas–Giannoni–Schmit criterion (Bohigas et al., 1984), genuine GOE requires a chaotic, strongly-mixed spectrum, which an integrable relaxation oscillator is not.
- **Retained (real).** The four-province compiled dataset, and the *superposition* result: pooling intervals across many independent sources drives the spacing ratio toward Poisson ($\langle r \rangle \rightarrow 0.386$). This is the standard superposition theorem for point processes (Daley & Vere-Jones, 2003) and is unaffected by the correction.
- **Added.** The shuffle (permutation) test on every series; the CV of the pooled spacings and an i.i.d. gamma renewal-clock comparison; a closed-data control showing the apparent

negative “rigidity” is a normalization artifact; and an age-error Monte Carlo showing the point estimate is unresolvable. New code: `code/audit_scalar_clock.py`.

- **Basis.** The methodological diagnoses are established in the scalar-clock audit note, Zenodo <https://doi.org/10.5281/zenodo.20980670>.
- **Preserved.** Version 1 (text, figures, results) is preserved in the git history (tag v1.0), in `paper/archive/v1/`, and as the prior Zenodo version. License unchanged (MIT).

2 Introduction

Ore deposits form in pulses. A single porphyry copper system, granite-related W–Sn deposit, or orogenic gold camp typically records several discrete mineralization stages (Sillitoe, 2010; Chelle-Michou et al., 2017). Version 1 of this paper asked whether the timing of those pulses carries the level-repulsion signature of random matrix theory (Wigner, 1957; Mehta, 2004), motivated by a *charge-and-release hypothesis*: each pulse depletes the local metal, fluid, and heat budget, which must re-accumulate before the next, imposing a minimum recurrence interval. Version 1 reported high spacing ratios in four provinces and read them as GOE/GUE repulsion.

That reading conflated two physically distinct situations. The spacing ratio $\langle r \rangle$ (Oganesyan & Huse, 2007; Atás et al., 2013) is *two-ended*. A value well above the Poisson expectation (0.386) can arise from (i) genuine sequential level repulsion, in which the position of each level depends on its neighbours (spectral rigidity), or (ii) a narrow *marginal* distribution of spacings with *no* sequential structure—a “scalar clock.” A relaxation oscillator that fires at a roughly fixed recurrence time (a charge-and-release reservoir, a dripping tap, a geyser) is exactly case (ii): its intervals cluster around a characteristic value, giving a high $\langle r \rangle$, while successive intervals are statistically independent. The two cases are distinguished by the *shuffle test*: randomly permuting the spacings destroys any sequential structure while preserving the marginal. Genuine repulsion is degraded by shuffling (the rigid spectrum has $\text{obs} < \text{shuf}$); a scalar clock is shuffle-invariant ($\text{obs} \approx \text{shuf}$). Version 1 never ran this test.

The deeper point is structural. By the Bohigas–Giannoni–Schmit criterion (Bohigas et al., 1984), GOE statistics are the fingerprint of a *chaotic, strongly-mixed* quantum (or wave) spectrum. A classical relaxation oscillator is integrable, not chaotic; its inter-event times form a renewal process. A charge-and-release ore reservoir is therefore, on physical grounds, a scalar renewal clock and *cannot* be a GOE spectrum, whatever the numerical value of $\langle r \rangle$. This erratum makes that argument quantitative on the original data.

3 Data and methods

3.1 Datasets (unchanged from v1)

We use the same compiled multi-stage absolute ages for four provinces: (1) *Andean porphyry copper* (subduction; 11 districts, Re–Os/U–Pb/Ar–Ar) (Maksaev et al., 2004; Romero et al., 2011; Deckart et al., 2013); (2) *South China Nanling W–Sn* (intracontinental granite; 15 deposits) (Mao et al., 2017; Zhou et al., 2016); (3) *Tethyan porphyry copper* (collisional; 9 districts across Tibet, Iran, Pakistan) (Hou et al., 2009; Yang et al., 2017); (4) *orogenic gold* (metamorphic, non-magmatic; 10 deposits across the Yilgarn, Abitibi, and West African cratons) (Vielreicher et al., 2010; Pelletier et al., 2022; Oberthür et al., 1998). Each district/deposit is treated as one source. These are representative literature ages, not a systematic census.

What the within-deposit “intervals” actually are. It is important to be explicit. Within a deposit, the intervals are differences between ages obtained by *different geochronometers* (U–Pb xenotime, U–Pb monazite, Ar–Ar muscovite, Re–Os sulfide) on what is often a single mineralizing episode. For Golden Mile, for example, the four ages 2675, 2655, 2645, 2630 Ma give “intervals” of 20, 10, 15 Myr—comparable to the individual ± 4 –5 Myr age uncertainties. The quantity version 1 analysed is therefore dominated by inter-method geochronological scatter, not by resolved province-scale recurrence.

3.2 RMT spacing ratio and the diagnostics v1 omitted

For each source we sort the ages, take inter-event intervals $s_i = x_{i+1} - x_i$, normalize each source to unit mean, and pool. We compute the unfolding-free consecutive spacing ratio

$$\langle r \rangle = \left\langle \frac{\min(s_i, s_{i+1})}{\max(s_i, s_{i+1})} \right\rangle, \quad (1)$$

with reference values Poisson 0.386, GOE 0.531, GUE 0.603 (Oganesyan & Huse, 2007; Atás et al., 2013). To this we add four diagnostics:

1. **Shuffle test.** Randomly permute the pooled spacings (preserving the marginal, destroying the sequence) and recompute $\langle r \rangle$. We report $\text{obs} - \text{shuf}$. A scalar clock gives $\text{obs} \approx \text{shuf}$; genuine GOE gives a small negative $\text{obs} - \text{shuf}$ (spectral rigidity), resolvable only for $n \gtrsim 300$.
2. **CV and gamma renewal clock.** We compute $\text{CV} = \sigma/\mu$ of the pooled spacings and compare $\langle r \rangle$ to that of a zero-repulsion i.i.d. gamma renewal process of the same CV (shape $k = \text{CV}^{-2}$), sampled at the same n .
3. **Closed-data control.** We push a zero-repulsion i.i.d. gamma null through the *same* within-deposit mean-normalization ($s_i \rightarrow s_i/\bar{s}$ per deposit) and pooling, and measure its

obs–shuf. Mean-normalization forces $\sum_i s_i = \text{const}$ per deposit, a closed/compositional constraint that induces negative correlation by construction.

4. **Age-error Monte Carlo.** We jitter every age within its quoted 1σ error (Gaussian), recompute the pooled $\langle r \rangle$, and report its distribution.

All diagnostics are implemented in `code/audit_scalar_clock.py`; the original pooled $\langle r \rangle$ is reproduced by `code/metallogeny_rmt.py`.

4 Results

4.1 The spacing ratios are reproduced—and are shuffle-invariant

We reproduce the version 1 spacing ratios exactly (Table 1). But the shuffle test, absent from version 1, removes the level-repulsion reading: at every province $\text{obs} \approx \text{shuf}$ (Figure 1b). For orogenic gold, $\langle r \rangle_{\text{obs}} = 0.678$ while $\langle r \rangle_{\text{shuf}} = 0.709$, i.e. $\text{obs} - \text{shuf} = -0.031$; the others are $+0.040$ (Andean), -0.001 (Nanling), $+0.074$ (Tethyan). There is no consistent negative rigidity. The high spacing ratios are a property of the *marginal* spacing distribution, not of the sequence.

Table 1: Audit of the four provinces. The version 1 spacing ratios are reproduced exactly, but are shuffle-invariant (column obs–shuf), are reproduced by a zero-repulsion gamma clock of the same CV, and are not resolvable within age error (MC range). n is the number of pooled within-source intervals.

Province (setting)	n	$\langle r \rangle_{\text{obs}}$ (=v1)	obs–shuf (shuffle)	CV	gamma clock $\langle r \rangle$ (95% band)	age-error MC $\langle r \rangle$ [2.5–97.5%]
Andean porphyry Cu (subduction)	40	0.601	+0.040	0.545	0.58 [0.49, 0.67]	[0.37, 0.56]
Nanling W–Sn (intracontinental)	36	0.574	−0.001	0.578	0.56 [0.47, 0.66]	[0.35, 0.57]
Tethyan porphyry Cu (collision)	21	0.712	+0.074	0.509	0.60 [0.48, 0.71]	[0.37, 0.65]
Orogenic gold (<i>non-magmatic</i>)	22	0.678	−0.031	0.310	0.73 [0.63, 0.81]	[0.39, 0.67]

4.2 A zero-repulsion clock of the same CV reproduces every value

The spacing ratio is fixed by the CV of the marginal distribution. A zero-repulsion i.i.d. gamma renewal process with the same CV, sampled at the same n , brackets the observed $\langle r \rangle$ in all four provinces (Table 1; Figure 1a): every province sits inside the finite- n 95% band of the scalar clock. The orogenic-gold case is the clearest—its low CV (0.31) is its high $\langle r \rangle$ (0.68); no repulsion is needed or implied.

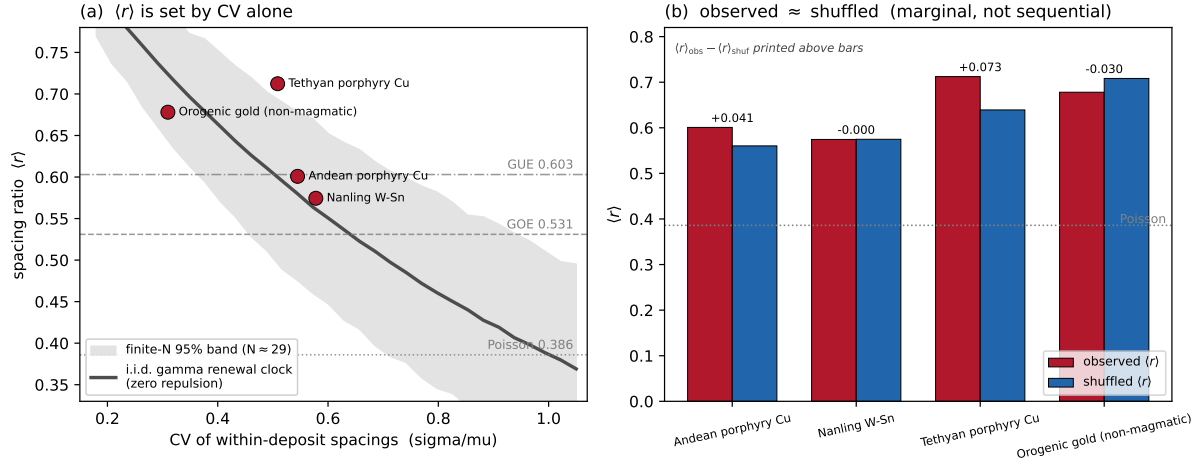


Figure 1: (a) The spacing ratio is set by the CV alone. The solid curve is $\langle r \rangle$ for a zero-repulsion i.i.d. gamma renewal clock as a function of CV, with the finite- n 95% band shaded; all four provinces sit inside it. (b) Observed vs shuffled $\langle r \rangle$. $\text{obs} \approx \text{shuf}$ at every province ($\text{obs} - \text{shuf}$ printed above the bars): the signal is in the marginal distribution, not the sequence.

4.3 The apparent “rigidity” is a closed-data artifact

The one province with a notable negative $\text{obs} - \text{shuf}$ is orogenic gold (-0.031). This is not spectral rigidity. Pushing a *zero-repulsion* i.i.d. gamma null through version 1’s own within-deposit mean-normalization and pooling yields $\text{obs} - \text{shuf} = -0.028 \pm 0.021$ for the gold configuration—statistically identical to the observed -0.031 . Mean-normalizing each deposit’s intervals to unit mean forces their sum to be constant, a closed (compositional) constraint that induces a spurious negative lag-1 correlation. The negative value is manufactured by the construction, not by the data.

4.4 The point estimate is geochronologically unresolvable

Jittering each age within its quoted 1σ error and recomputing the pooled $\langle r \rangle$ produces broad distributions (Table 1; Figure 2c). For orogenic gold the central 95% of the Monte Carlo spans $\langle r \rangle \in [0.39, 0.67]$ —from the Poisson value to above GUE—with a median of 0.53. The version 1 reported value (0.678) sits at the very top edge of that envelope. At $n = 21\text{--}40$, far below the $n \gtrsim 300$ needed to separate a scalar clock from GOE by rigidity, the spacing ratio simply cannot bear the interpretive weight version 1 placed on it.

There is a deeper, instrument-level reason the sequential reading was never admissible. Genuine level repulsion is a property of the *ordering* of closely spaced levels, so detecting it demands an absolute time resolution finer than the spacings being ordered. The chronometers underlying these compilations—U–Pb on zircon and xenotime, $^{40}\text{Ar}/^{39}\text{Ar}$ on muscovite, Re–Os on arsenopyrite and molybdenite—carry intrinsic 1σ uncertainties of order $\pm 1\text{--}5$ Ma, which are comparable to or larger than the within-source intervals themselves. Any micro-

scale sequential rigidity that a Wigner–Dyson spectrum would require is therefore convolved away by the dating errors before it could be measured: the analytical resolution available in ore geochronology is, on its face, insufficient to support a level-repulsion statistic. The high $\langle r \rangle$ that does survive this smearing is exactly what a narrow marginal (a scalar clock) produces, because $\langle r \rangle$ from a narrow CV is robust to jitter while sequential rigidity is not.

4.5 What survives: superposition $\rightarrow$ Poisson

The superposition result is unaffected and we retain it. When intervals from many independent sources are pooled *without* source isolation, the spacing ratio collapses toward Poisson (Figure 2d): mixed $\langle r \rangle = 0.47$ (Andean), 0.31 (Nanling), 0.39 (Tethyan), 0.43 (gold), against single-source values of 0.57–0.71. This is the standard superposition theorem—the sum of many independent renewal processes tends to Poisson (Daley & Vere-Jones, 2003)—and it is a genuine, mechanism-independent feature of the data. It says nothing about level repulsion; it says that mixing erases the marginal clock of each source.

Removing the spurious single-source GOE claim does not weaken this result; it strengthens it. The superposition collapse never depended on the local statistic being GOE—only on each source carrying *some* non-Poisson marginal clock, which the scalar-clock reading supplies just as well. With the pseudo-repulsion artifact cleared away, the one robust, scale-independent statement the pipeline supports stands on firmer ground: independent metallogenic sources, superposed, lose their individual rhythms and the pooled timing relaxes to a memoryless Poisson process. That is the finding worth carrying forward.

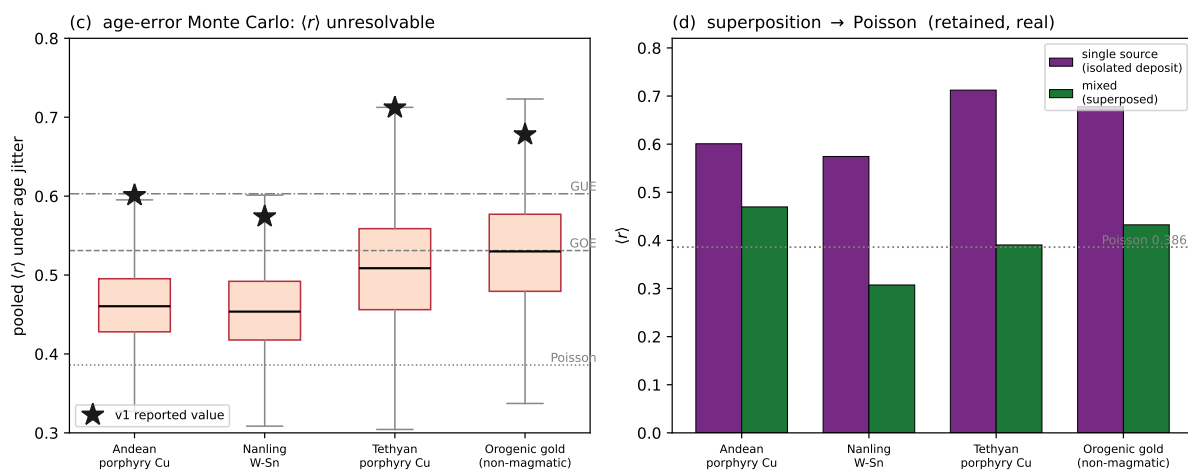


Figure 2: (c) Age-error Monte Carlo. Box plots of the pooled $\langle r \rangle$ when ages are jittered within their quoted 1σ errors; stars mark the version 1 reported values. The estimates float across the Poisson–GUE range; the v1 gold value sits at the top edge. (d) The retained, real result: single (isolated-source) vs mixed (superposed) $\langle r \rangle$. Superposition drives the local statistic toward Poisson (0.386).

5 Discussion

5.1 Scalar clock versus level repulsion

The corrected picture places these systems firmly at the scalar-clock end of the diagnostic axis. A high $\langle r \rangle$ that is shuffle-invariant, CV-determined, and reproduced by an i.i.d. renewal clock is the signature of a narrow marginal spacing distribution—a system with a characteristic recurrence time and independent successive intervals. That is precisely what a charge-and-release reservoir is: a relaxation oscillator. It is the same object as a dripping tap, a geyser, a glacial jökulhlaup cycle, or a nucleosome spacing ruler—all of which give high $\langle r \rangle$ for the same reason and none of which is a GOE spectrum. Genuine level repulsion, by contrast, requires a chaotic, strongly-mixed spectrum (open d/f shells, heavy nuclei, hydrogen in a strong magnetic field) per the Bohigas–Giannoni–Schmit criterion (Bohigas et al., 1984), and shows resolvable spectral rigidity ($\text{obs} < \text{shuf}$) at large n . The ore data show none of this.

The distinction is one of dynamical class, not of degree. A charge-and-release reservoir has a single effective degree of freedom—an accumulating budget that discharges past a threshold—so its dynamics live on a classical limit cycle and its inter-event statistics are those of a one-dimensional renewal process. Wigner–Dyson statistics, on the other hand, are the fingerprint of a high-dimensional, strongly-coupled, non-integrable system whose levels genuinely interact. Reading a metallogenic clock as GOE therefore mistakes a classical, effectively isolated relaxation oscillation for quantum-grade strong coupling—it assigns a one-degree-of-freedom threshold process to a universality class reserved for chaotic many-body spectra. The general lesson for applied RMT is that a high spacing ratio is a necessary but not sufficient condition for level repulsion: without the shuffle test, a CV check, and a renewal-clock null, any narrow-marginal recurrence will masquerade as a repelling spectrum.

5.2 Why the non-magmatic “control” does not rescue the claim

Version 1’s strongest argument was that orogenic gold—non-magmatic—shows the same high $\langle r \rangle$, supposedly cutting off an “inherited from magma” explanation and implying intrinsic repulsion. The argument fails because the premise (that the high $\langle r \rangle$ is repulsion) is false. Orogenic gold has the same high $\langle r \rangle$ as the magmatic provinces because it is the same kind of object—a scalar renewal clock (here, a low-CV cluster of multi-method ages on a few mineralizing episodes), not because it shares an intrinsic repulsion mechanism. The cross-cratonic consistency of the gold $\langle r \rangle$ is consistency of a scalar clock, not of level repulsion.

5.3 What can still be said

Two statements remain defensible. First, individual ore systems behave as scalar charge-release clocks: their pulse intervals cluster around a characteristic recurrence time, consistent with

reservoir recharge dynamics, but with no evidence of sequential level repulsion at present resolution. Second, superposition of many sources randomizes timing toward Poisson, so any clock signature is a single-source phenomenon. Neither statement invokes random-matrix universality, and the paper no longer claims it.

5.4 Limitations

Beyond the corrected interpretation: sample sizes are small ($n = 21\text{--}40$ pooled intervals); dating methods are heterogeneous and the within-deposit intervals are dominated by inter-method scatter; ages are representative literature values, not a census; and the geochronological errors alone make the spacing ratio unresolvable. A genuine test of recurrence structure would need many tens of resolved, single-method pulse ages within one system—currently unavailable.

6 Conclusions

We retract the version 1 claim that single ore-forming systems show GOE/GUE level repulsion. The high spacing ratios are reproduced exactly but are shuffle-invariant, CV-determined, reproduced by a zero-repulsion renewal clock, contaminated by a closed-data normalization artifact in their apparent “rigidity,” and unresolvable within age error. A charge-and-release reservoir is a relaxation oscillator—a scalar renewal clock—which by the Bohigas–Giannoni–Schmit criterion is structurally not a GOE spectrum. What survives is the superposition result: pooling independent sources drives the timing toward Poisson. We keep the data and the superposition analysis, and reframe single ore systems as scalar charge-release clocks rather than level-repulsive spectra. The correction follows the scalar-clock audit, Zenodo <https://doi.org/10.5281/zenodo.20980670>.

Code and Data Availability

All code (including the new `code/audit_scalar_clock.py`), compiled datasets, and figures: <https://github.com/Ruqing1963/metallogeny-rmt>. Version 1 is preserved in the git history (tag `v1.0`), in `paper/archive/v1/`, and as the prior Zenodo version. Methodological basis of the correction: Zenodo <https://doi.org/10.5281/zenodo.20980670>.

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